Algorithms for Data Science
Cheatsheet for Exercises 3

Master Theorem. *Let* $\boldsymbol{T(n) = aT(n/b) + f(n)}$ *with constants* $a > 0$, $b > 1$, *and* $f(n)$ *being a nondecreasing function. Let* $c = \log_b a$.

- ***(Case A)*** *If* $f(n) = O(n^{c'})$ *for some* $c' < c$, *then* $\boxed{T(n) = \Theta(n^c)}$.
- ***(Case B)*** *If* $f(n) = \Theta(n^c)$, *then* $\boxed{T(n) = \Theta(n^c \log n)}$.
- ***(Case C)*** *If* $f(n) = \Omega(n^{c'})$ *for some* $c' > c$, *then*[(*)] $\boxed{T(n) = \Theta(f(n))}$.

[(*)]*Note: for Case C,* $f(n)$ *must also satisfy the* regularity condition*: there is a constant* $k < 1$ *such that for all sufficiently large* n, *we have* $af(n/b) \leq kf(n)$. *However, in our course, you don't have to learn it.*

It is very easy to decide which case holds if $f(n)$ *is a polynomial:*

Master Theorem "Lite"
Compute $c = \log_b a$ as above. If $f(n) = \Theta(n^d)$ for some constant number d such that

- $\ldots d < c$, then Case A holds.
- $\ldots d = c$, then Case B holds.
- $\ldots d > c$, then Case C holds.

Note: the regularity condition holds automatically for Case C if $f(n) = \Theta(n^d)$.

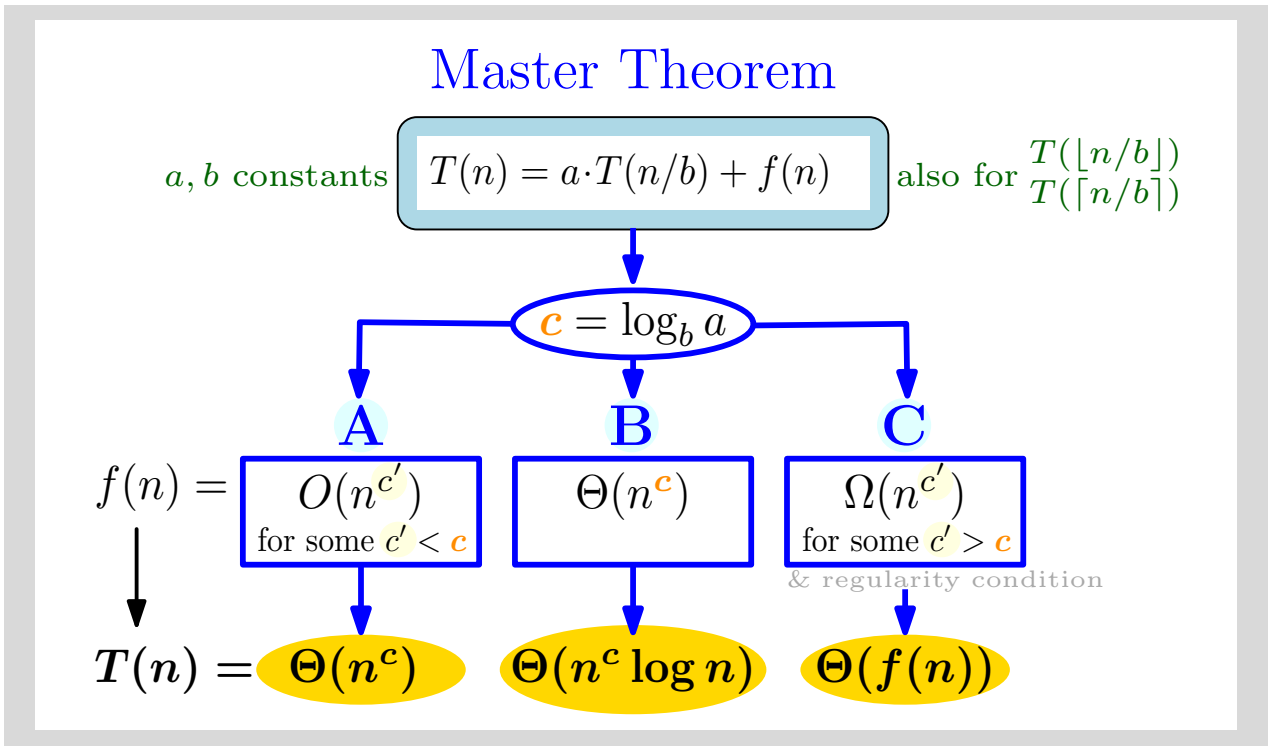
Nice to Know. *If* $T(n) = aT(\lfloor n/b \rfloor) + f(n)$ *or* $T(n) = aT(\lceil n/b \rceil) + f(n)$, *then we can simplify and write* $T(n) = aT(n/b) + f(n)$. *This simplification does not change the solution to* $T(n)$.

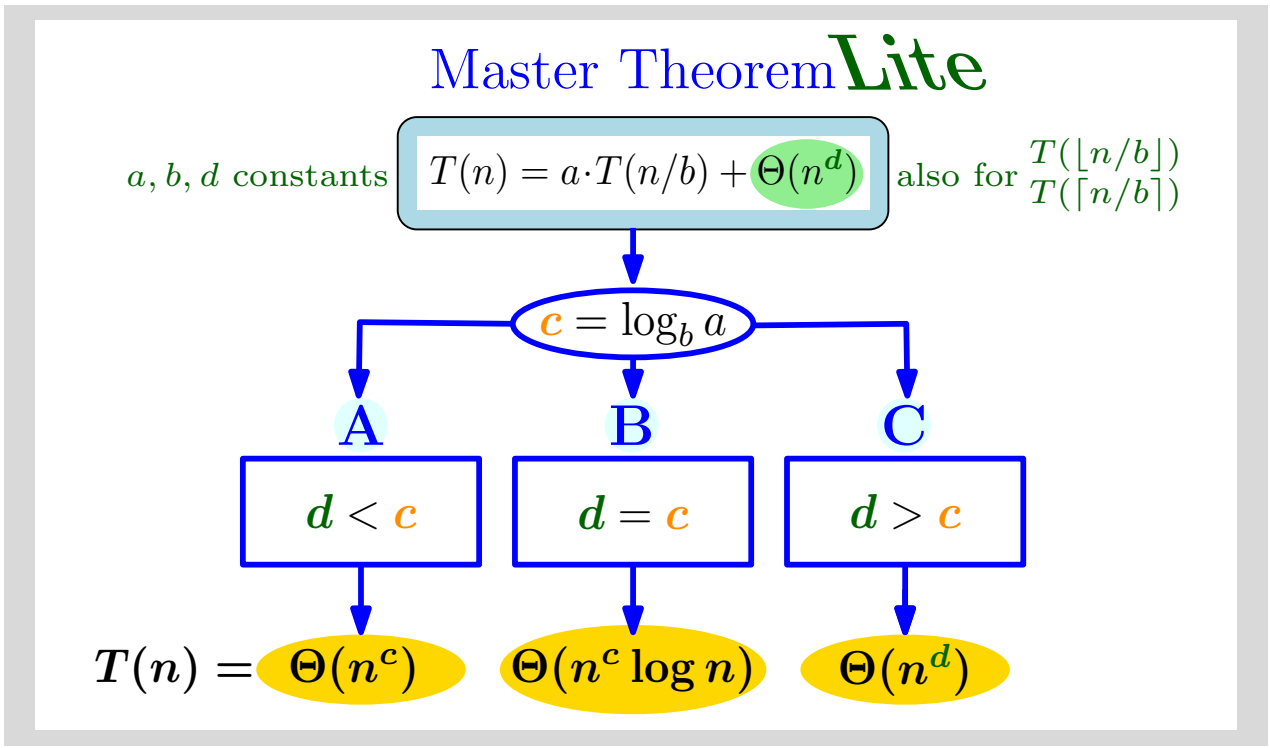
Notation. Recall that

- $\lfloor\ \rfloor$ is the *floor* function that rounds a given number *down* to the next integer.
  Examples: $\lfloor 1.5 \rfloor = 1$ *and* $\lfloor 4 \rfloor = 4$.
- $\lceil\ \rceil$ is the *ceiling* function that rounds a given number *up* to the next integer.
  Examples: $\lceil 1.5 \rceil = 2$ *and* $\lceil 4 \rceil = 4$.

Summation Rules.

1. (First n integers) $1 + \cdots + n = \sum_{i=1}^{n} i = n(n+1)/2 = \Theta(n^2)$
2. (First n power of twos) $1 + 2 + 2^2 + \cdots + 2^n = \sum_{i=0}^{n} 2^i = 2^{n+1} - 1 = \Theta(2^n)$


Master Theorem
a, b constants
$T(n) = a \cdot T(n/b) + f(n)$
also for $T(\lfloor n/b \rfloor)$ $T(\lceil n/b \rceil)$
$c = \log_b a$
A
B
C
$f(n) =$
$O(n^{c'})$ for some $c' < c$
$\Theta(n^c)$
$\Omega(n^{c'})$ for some $c' > c$
& regularity condition
$T(n) =$
$\Theta(n^c)$
$\Theta(n^c \log n)$
$\Theta(f(n))$


Master Theorem Lite
a, b, d constants
$T(n) = a \cdot T(n/b) + \Theta(n^d)$
also for $T(\lfloor n/b \rfloor)$ $T(\lceil n/b \rceil)$
$c = \log_b a$
A
B
C
$d < c$
$d = c$
$d > c$
$T(n) =$
$\Theta(n^c)$
$\Theta(n^c \log n)$
$\Theta(n^d)$